Order, Consistency and Information States: A Requirement on the Semantics of Epistemic Modals

Abstract

Epistemic "might" shows an order effect in discourse. Once the information that it is not raining has been added, it is no longer consistent with the information that it might be raining. When the two are added in the opposite order, no problem arises. This paper reconstructs how dynamic semantics explains that contrast and argues that the explanation supports a condition of adequacy: an adequate semantics of epistemic modals must make modal evaluation sensitive to information states, not only to worlds. The argument has two premises. The first is the dynamic treatment of conjunction as sequential and of the might-operator as a consistency test. The second is the order contrast itself. Together they show that changing the sequence of information changes the state against which a might-claim is tested, and so changes whether the claim passes that test. The conclusion is a demand on adequate accounts. It does not say that worlds play no role. The paper does not independently establish the premises it relies on, and a single example does not show that every epistemic modal behaves this way.

1. Introduction

What must a semantics of epistemic modals take into account when it evaluates a claim such as "it might be raining"? This paper argues for a condition that any adequate answer must meet: an adequate semantics of epistemic modals must make modal evaluation sensitive to information states, not only to worlds. The condition adds a requirement and takes nothing away. It does not say that worlds play no role in modal evaluation. It says that worlds alone are not enough, so information states must also matter. It is also a condition of adequacy, not a report. It says what an account must do, not that some account meeting the condition has already been established.

The argument has two premises, both taken from a dynamic-semantic source. The first is a pair of semantic commitments: dynamic semantics gives conjunction a sequential interpretation and interprets the might-operator as a consistency test (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). The second is an order contrast in discourse. After information that it is not raining is added, it is no longer consistent with information that it might be raining; in the opposite order there is no problem, so order matters in discourse (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). The inference is from a sign. The order effect is treated as evidence about what modal evaluation must be sensitive to, and the commitments explain why it counts as evidence of that.

The paper proceeds as follows. Section 2 sets out the argument in three steps. It first separates the two commitments and states what each says and does not say (§2.1). It then states the order contrast and fixes the scope of each half (§2.2). Finally, it shows how the commitments make the contrast bear on

modal evaluation, and in which direction the support runs (§2.3). Section 3 takes up the two critical questions the inference has to answer: whether the order contrast is reliable evidence, and whether the commitments really support the conclusion. It also addresses two misreadings that the answers expose. Section 4 states the limits of the support. Section 5 places the argument relative to a bounded search of recent literature, and Section 6 concludes.

The aim is narrow. The paper does not propose a new semantics, and it does not defend the dynamic commitments on independent grounds. Its contribution is to make explicit how two specific commitments and one specific contrast together support a requirement on adequate accounts, and to state how far that support reaches. Section 4 states the boundaries of the result where they matter, rather than repeating them throughout.

2. Argument

2.1 Sequential conjunction and consistency testing

The first premise states that dynamic semantics gives conjunction a sequential interpretation and interprets the might-operator as a consistency test (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). This premise contains two commitments, and they concern different expressions. The first concerns conjunction; the second concerns the might-operator. Keeping them apart shows what each contributes to the explanation in §2.3. It also keeps either feature from being assigned to the wrong operator: sequentiality is not claimed for "might", and consistency testing is not claimed for conjunction.

The first commitment says only how conjunction is handled: it is interpreted sequentially, rather than without regard to the order of its parts. As stated, it sets out no further stages or outcomes, gives no rule for how a particular conjunct changes an information state, and makes no claim about what follows from any particular ordering. The second commitment treats the might-operator as a consistency test. It too comes without further detail about how the test operates. What it does fix is how modal evaluation relates to information: the question for a might-claim is whether it remains consistent with the information state against which it is tested (Groenendijk, Stokhof & Veltman 1996, PDF p. 18).

Together, the two commitments shape how epistemic-modal claims are framed. Sequence is part of how conjunction is interpreted, and consistency with information is part of how "might" is interpreted. A might-claim that occurs in a sequence is therefore tested against an information state, and which state that is depends on what has already been added. The scope of this pairing is limited. It adds no further semantic rule. It makes no claim about other properties of conjunction or about other modal operators. And it does not by itself amount to a general claim about all epistemic modals (Groenendijk, Stokhof & Veltman 1996, PDF p. 18).

2.2 The not-raining order effect

The second premise concerns the order in which information is added in discourse. In one order, information that it is not raining is added before information that it might be raining. The stated result is that once the not-raining information has been added, it is no longer consistent with the information that it might be raining. In the opposite order, there is no problem (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). The contrast is therefore not between two pieces of information considered apart from sequence. It is between two orders of adding the same two pieces of information, and the two orders have different outcomes.

Each half of the contrast has a narrow scope. The first half ties the consistency result to the order in which the not-raining information comes first. It does not say that the same result holds whichever piece of information is added first. Nor does it support a broader conclusion about every possible order, every kind of information or every epistemic modal. The example concerns only the information that it is not raining and the information that it might be raining. The second half says only that the reverse sequence presents no problem, and it specifies no further outcome for that sequence. Its job is to mark the difference between the two orders. The contrast needs both halves, and neither should be read more broadly than it is stated (Groenendijk, Stokhof & Veltman 1996, PDF p. 18).

From this contrast, the source concludes that order matters in discourse (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). "Order matters" is a name for the contrast. It adds no further outcome, introduces no other example and states no general rule about other information, other sequences or other modal expressions. Read narrowly, the claim is this: for the two specified pieces of information, whether the sequence presents a problem depends on which piece comes first (Groenendijk, Stokhof & Veltman 1996, PDF p. 18).

2.3 From the contrast to information-state sensitivity

The commitments are what make this contrast bear on modal evaluation. "Might" tests consistency on an information state, and conjunction is interpreted sequentially. Each addition of information therefore changes the state against which what follows is tested. Adding the same information in a different sequence changes whether the might-claim remains consistent when it is tested. When the not-raining information comes first, the state the might-information meets is no longer consistent with it. In the reverse order, no problem arises. Read through the commitments, the order effect is explained (Groenendijk, Stokhof & Veltman 1996, PDF p. 18).

The explanation also licenses the step from contrast to evidence. The contrast is about consistency after information has been added, not about an order-free relation between the two items. The two orders contain the same information and differ only in sequence. On the commitments, sequence matters because it fixes the information state against which the might-test is run. The outcome of the test therefore tracks that state. This is why the contrast counts as a sign that epistemic-modal evaluation is sensitive to information states, and not to worlds alone. It is the step on which the conclusion announced in §1 rests (Groenendijk, Stokhof & Veltman 1996, PDF p. 18).

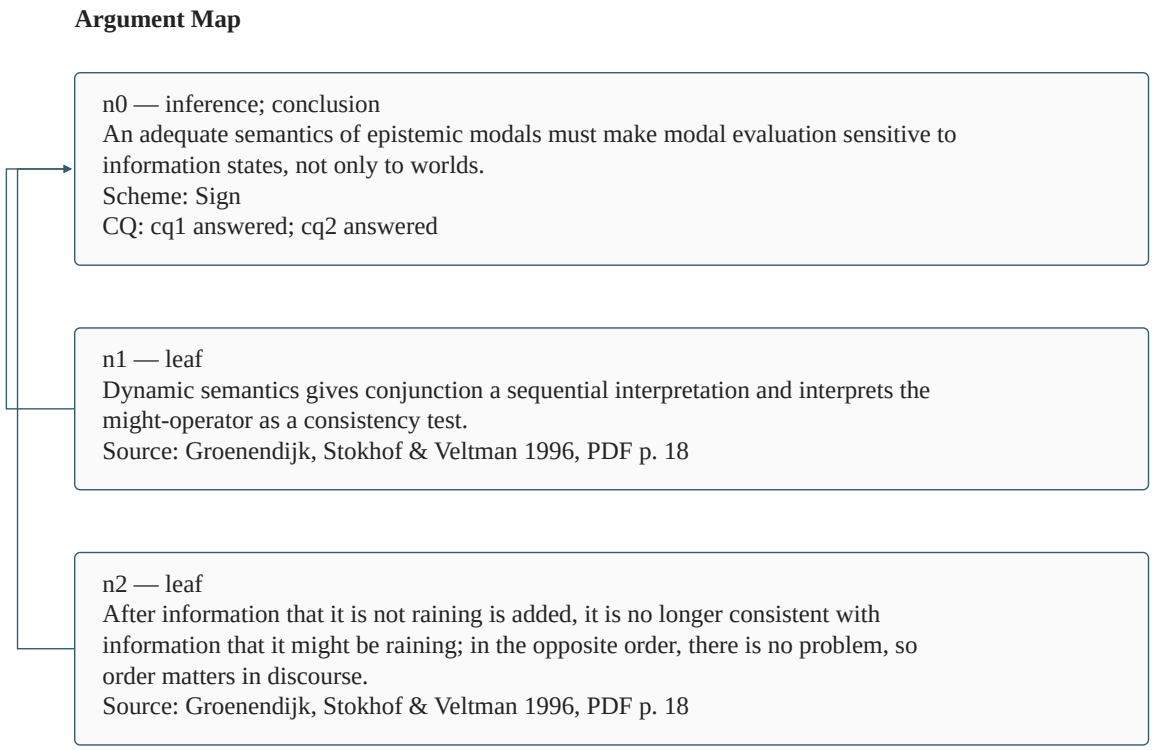


Figure 1. Route premises, inference schemes and conclusion. Arrows follow premise-to-conclusion links; CQ labels report recorded answers.

The support runs in one direction only: from the commitments and the order effect to the requirement on an adequate semantics. The requirement does not independently prove the commitments, and nothing in the argument uses it that way. The inference is therefore conditional. If conjunction is interpreted sequentially, "might" tests consistency, and the order contrast holds, then the contrast supports the requirement. The paper reports that conditional connection and its direction. It does not establish the antecedent independently (Groenendijk, Stokhof & Veltman 1996, PDF p. 18).

3. Objections and replies

The first critical question is whether the order contrast is reliable as evidence. One may doubt that a contrast drawn from a single pair of claims about rain can carry the weight the argument puts on it. The worry has force, because the argument rests on this one example. The reply is that the contrast is reliable in the role the argument gives it: as evidence that epistemic-modal evaluation is sensitive to information states. It is reliable in that role because it is explained. When conjunction updates information sequentially and "might" tests consistency, the difference between the two orders follows (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). The evidence is not a bare report that the two orders differ. It comes with an account of why the outcome changes with order.

The same answer records a limit, and the objection is right to press it. The order effect does not by itself establish that every epistemic modal behaves this way. The reply does not deny this. Instead, it points out that the conclusion was never a generalization over all epistemic modals. It is a demand on

an adequate semantics, backed by these commitments and this example, and it does not need the wider claim (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). The objection shows how the conclusion must be read. It does not show that the evidence fails.

The second critical question is whether the commitments support the conclusion at all. The commitments describe how dynamic semantics treats conjunction and "might". One might object that a requirement on any adequate semantics cannot be drawn from features of one framework without presupposing that framework, so the support would be only as strong as the framework. The reply has two parts. First, the commitments do support the conclusion. The might-operator is treated as a consistency test on an information state, and adding information in a different sequence changes whether the might-claim remains consistent. That supports information-state-sensitive evaluation rather than evaluation by worlds alone (Groenendijk, Stokhof & Veltman 1996, PDF p. 18).

Second, the objection correctly picks out the conditional character of the support, which §2.3 already made explicit. The argument does not establish independently that conjunction is sequential or that "might" tests consistency. What it establishes is that if those commitments and the order contrast hold, the contrast supports the requirement. The objection therefore marks the extent of the result rather than refuting it. Anyone who accepts the premises has reason to accept the requirement, and the paper claims no more than that.

A third objection concerns what the conclusion says. If evaluation must be sensitive to information states, it may seem that worlds drop out of modal evaluation altogether. This misreads the requirement. Its wording is additive: sensitivity to information states, *not only* to worlds. Nothing in the commitments or the contrast says that worlds have no role, and the argument draws no such consequence. The evidence bears on what must be added to modal evaluation, not on what must be taken out of it.

A fourth objection asks whether the contrast concerns modal evaluation or only discourse. The source sums up the contrast by saying that order matters in discourse (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). A fact about discourse order might seem to say nothing about how a might-claim is evaluated. The reply again turns on the commitments. On the might-commitment, what changes between the two orders is whether the might-claim remains consistent with the information state against which it is tested. Because the test is part of the interpretation of "might", a difference in its outcome is a difference in how the modal claim is evaluated (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). The link comes from the commitment, not from the contrast alone. This is why the argument needs both premises.

4. Limits

The first limit is generality. The example concerns two specified pieces of information, that it is not raining and that it might be raining, and the two orders of adding them. As the reply to the first objection grants, it does not by itself show that every epistemic modal behaves this way. The paper draws no conclusion about other sequences, other kinds of information or other modal expressions (Groenendijk, Stokhof & Veltman 1996, PDF p. 18).

The second limit is that the commitments are stated at a level of detail that restricts what can be derived from them. The sequential interpretation of conjunction comes without rules for how a particular conjunct changes an information state. The consistency test comes without further detail about how it operates (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). The argument therefore uses the commitments only to connect sequence with the information state against which "might" is tested. On this basis it cannot say in detail how updating proceeds, or predict outcomes for cases beyond the stated contrast.

The third limit is that the reconstruction is thinner than its source. The source presents the order contrast as explained, but the recorded support for that premise omits a supporting inference from the source (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). That inference is not among the premises reconstructed here, and the present argument does not rely on it. The explanation in §2.3 should therefore be read as using only the two stated commitments, not as reproducing the source's full derivation (Groenendijk, Stokhof & Veltman 1996, PDF p. 18).

Source Support

n0 ← Groenendijk, Stokhof & Veltman 1996, PDF p. 18
Stance: endorsed; attributed to Groenendijk, Stokhof and Veltman; role: cq_evidence

n0 ← Groenendijk, Stokhof & Veltman 1996, PDF p. 18
Stance: endorsed; attributed to Groenendijk, Stokhof and Veltman; role: cq_evidence

n1 ← Groenendijk, Stokhof & Veltman 1996, PDF p. 18
Stance: endorsed; attributed to Groenendijk, Stokhof and Veltman; role: global_premise

n2 ← Groenendijk, Stokhof & Veltman 1996, PDF p. 18
Stance: endorsed; attributed to Groenendijk, Stokhof and Veltman; role: global_premise

Figure 2. Frozen source support by route node, page citation, attribution and stance. Recorded support is not independent verification.

The fourth limit is that the premises are taken as given. Section 2.3 and the reply to the second objection set out the conditional form of the support. The point here is that this paper adds no independent evidence for the commitments or the contrast beyond the source in which they are stated (Groenendijk, Stokhof & Veltman 1996, PDF p. 18).

5. Related work and scope of the novelty claim

This section places the argument relative to a bounded literature search covering 2018 through 28 October 2026 (Literature-currency search 2026, 2018–2026 search). The search targeted literature on contextualism, relativism, expressivism, dynamic semantics, epistemic contradictions, probability operators, retraction and assessment sensitivity (Literature-currency search 2026, 2018–2026 search). It selected 25 recent works and obtained 17 exact full texts, plus three previously missing originals; eight selected recent works and 24 originals remained unavailable (Literature-currency search 2026, 2018–2026 search). It is a ranked, bounded search of indexes and public sources, not an exhaustive review (Literature-currency search 2026, 2018–2026 search). Its record notes that acquired files had not been extracted into the frozen evidence on which this paper's argument rests, and that material access gaps include older foundational chapters, complete books, the strongest 2024 safety/expressivism comparisons and recent empirical reviews (Literature-currency search 2026, 2018–2026 search). A failed search or an inaccessible paper establishes neither an open problem nor novelty (Literature-currency search 2026, 2018–2026 search). A later focused follow-up concerned a different question, the relation between worldly-information safety and retraction, and its verdict is not applied to the present thesis (Literature-currency search 2026, 2018-01-01 through 2026-10-28).

Within that search, the paper's thesis is partially anticipated by read works (Mandelkern 2025, PDF pp. 1, 2, 4, 12, 16, 17; §2–6, especially6.2 Order) (Močnik 2019, PDF pp. 1, 2; §abstract and introduction). Information-sensitive approaches to epistemic modals already exist, and the survey of dynamic and static options read for this assessment treats them alongside static alternatives (PDF pages 1, 2, 4, 12, 16 and 17, §§2–6; this is an author chapter manuscript dated 24 January 2025 that excerpts chapters 2 and 4 of a 2024 book, not the full book) (Mandelkern 2025, PDF pp. 1, 2, 4, 12, 16, 17; §2–6, especially6.2 Order). Alternatives in the read literature also question whether information sensitivity requires a separate semantic parameter or a nonclassical semantics (Mandelkern 2025, PDF pp. 1, 2, 4, 12, 16, 17; §2–6, especially6.2 Order) (Močnik 2019, PDF pp. 1, 2; §abstract and introduction). A pragmatic competition account of modal embedding contrasts, read in its proceedings version (PDF pages 1–2, abstract and introduction), holds that modal embedding is not automatically evidence that uniquely favors one semantic family (Močnik 2019, PDF pp. 1, 2; §abstract and introduction). A further relevant work, "Epistemic Contradictions Do Not Threaten Classical Logic", was available only as an abstract; it is named here as an access gap, not as a read text, and no claim about its content is relied on.

Using discourse order as a diagnostic is likewise established in existing work rather than introduced here (Mandelkern 2025, PDF pp. 1, 2, 4, 12, 16, 17; §2–6, especially6.2 Order). The same survey treats order effects explicitly, especially in §6.2, "Order" (Mandelkern 2025, PDF pp. 1, 2, 4, 12, 16, 17; §2–6, especially6.2 Order). It also states a modern limit on carrying them over: discourse-order effects do not by themselves establish within-sentence semantic order effects (Mandelkern 2025, PDF pp. 1, 2, 4, 12, 16, 17; §2–6, especially6.2 Order). Relatedly, a narrow static model that treats sentence sequences as conjunctions predicts no order contrast (Mandelkern 2025, PDF pp. 1, 2, 4, 12, 16, 17; §2–6, especially6.2 Order). Rejecting static semantics in general, however, would require

more, since pragmatic explanations and embedded symmetry would have to be addressed (Mandelkern 2025, PDF pp. 1, 2, 4, 12, 16, 17; §2–6, especially6.2 Order) (Močnik 2019, PDF pp. 1, 2; §abstract and introduction). The present paper does not attempt that wider rejection.

A read critical notice, consulted as an author manuscript (PDF pages 1–2), directly compares dynamic semantics with dynamic propositionalism (Willer 2023, PDF pp. 1, 2; §The Plot; critical notice of Stojnić). No neutral resolution of that debate is inferred here. Measured against this literature, the paper's contribution is modest. It reconstructs how two stated commitments and one order contrast support an adequacy condition already partly on the table, and in that respect it is partially anticipated by the survey and the pragmatic account cited above (Mandelkern 2025, PDF pp. 1, 2, 4, 12, 16, 17; §2–6, especially6.2 Order) (Močnik 2019, PDF pp. 1, 2; §abstract and introduction). The bounded search provides no basis for a novelty claim, and none is made (Literature-currency search 2026, 2018–2026 search).

6. Conclusion

This paper has argued that an adequate semantics of epistemic modals must make modal evaluation sensitive to information states, not only to worlds. The support comes from two premises. First, dynamic semantics gives conjunction a sequential interpretation and interprets "might" as a consistency test. Second, once the not-raining information has been added, it is no longer consistent with the information that it might be raining, while the opposite order presents no problem (Groenendijk, Stokhof & Veltman 1996, PDF p. 18). Read through the commitments, changing the order changes the information state against which the might-claim is tested, and so changes whether the claim remains consistent. That is why the contrast counts as evidence about what modal evaluation tracks.

The result is a condition of adequacy, not a theory, and its reach is limited accordingly. It leaves worlds a role. It depends on premises the paper does not independently establish. And the single example it rests on does not show that every epistemic modal behaves this way. What it does establish is the direction of support: from the commitments and the order effect to a requirement that any account of epistemic modals must meet (Groenendijk, Stokhof & Veltman 1996, PDF p. 18).

References

Groenendijk, Jeroen; Stokhof, Martin; Veltman, Frank (1996). Coreference and Modality. In S. Lappin (ed.), The Handbook of Contemporary Semantic Theory, Blackwell. pp. 179-213.

Matthew Mandelkern (2025). Dynamic semantics for epistemic modality.

Maša Močnik (2019). Embedded Epistemic Modals Pragmatically. https://doi.org/10.18148/sub/2019.v23i2.606.

Malte Willer (2023). Dynamic semantics versus dynamic propositionalism. https://doi.org/10.1080/0020174x.2023.2174735.

Literature-currency record (2026). Bounded 2018–2026 search; verified page/section references and access limits retained in the accompanying publication receipt.